

Food Insecurity and Youth Depression Risk in the United States

A National Epidemiologic Analysis Using NHANES 2005-2022

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Repository: <https://github.com/bobaoxu2001/Food-insecurity>

Indicator	Value
Combined sample (all ages)	82,123
Youth sample (12-19 years)	11,659
Complete-case analytic sample	2,081
Depression prevalence in food-secure youth	7.0%
Depression prevalence in food-insecure youth	17.3%
Adjusted OR for food insecurity	3.04 (95% CI 2.13-4.34), p <0.0001

Research question: Does household food insecurity increase the risk of depressive symptoms among adolescents and youth in the United States?

1. Background and Literature Context

Food insecurity is a major social determinant of health linked to both physical and psychological burden. Adolescents exposed to unstable household food access may face chronic stress, social strain, and reduced diet quality, which can collectively elevate depression risk.

Prior epidemiologic evidence has reported consistent associations between food insecurity and poor mental health outcomes. This report evaluates whether those patterns persist in a pooled, nationally representative U.S. youth sample.

2. Data and Study Design

Data were drawn from the National Health and Nutrition Examination Survey (NHANES), pooled from 2005-2006 through 2021-2022 (8 cycles).

- Modules merged: DEMO, DPQ (PHQ-9), FSQ, and BMX.
- Target population: youth aged 12-19 years.
- Primary outcome: depressive symptoms (PHQ-9 ≥ 10).
- Primary exposure: food insecurity (low/very low vs full/marginal).
- Food-security variable harmonization: FSDHH when available, FSDAD otherwise.
- Covariates: age, sex, income-to-poverty ratio, and BMI.

3. Statistical Analysis

Weighted logistic regression was fit as: Depression $\sim$ Food Insecurity + Age + Sex + Income + BMI. MEC weights were pooled across cycles and normalized to preserve weighted point estimates while keeping inference stable.

4. Results

Food insecurity showed a strong positive association with depressive symptoms (OR 3.04, 95% CI 2.13-4.34, $p < 0.0001$). Weighted prevalence was 7.0% among food-secure youth and 17.3% among food-insecure youth.

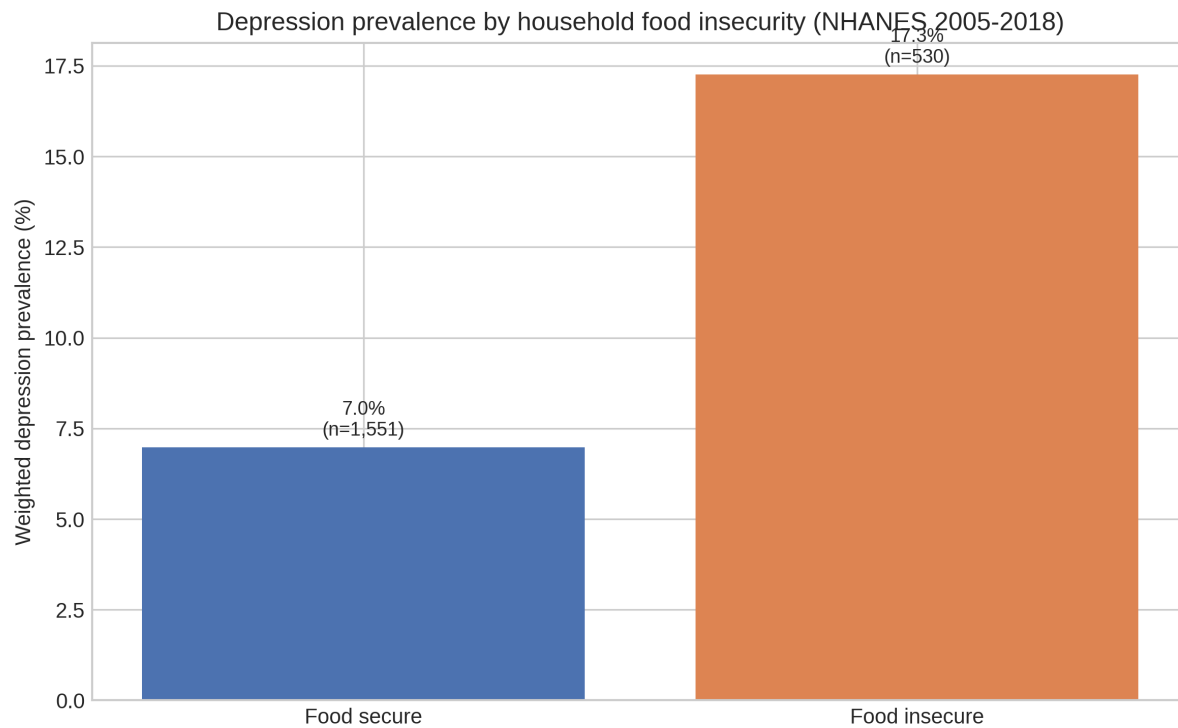


Figure 1. Weighted prevalence of depressive symptoms by household food insecurity status.

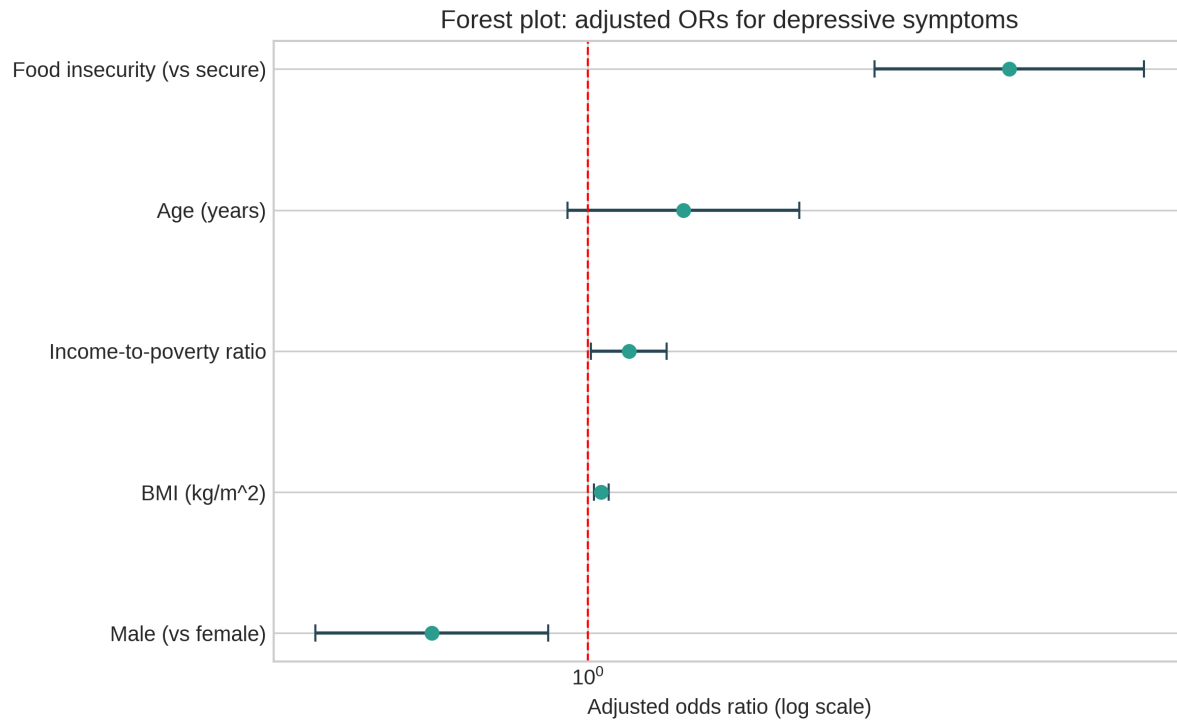


Figure 2. Forest plot of adjusted odds ratios from the multivariable weighted logistic model.

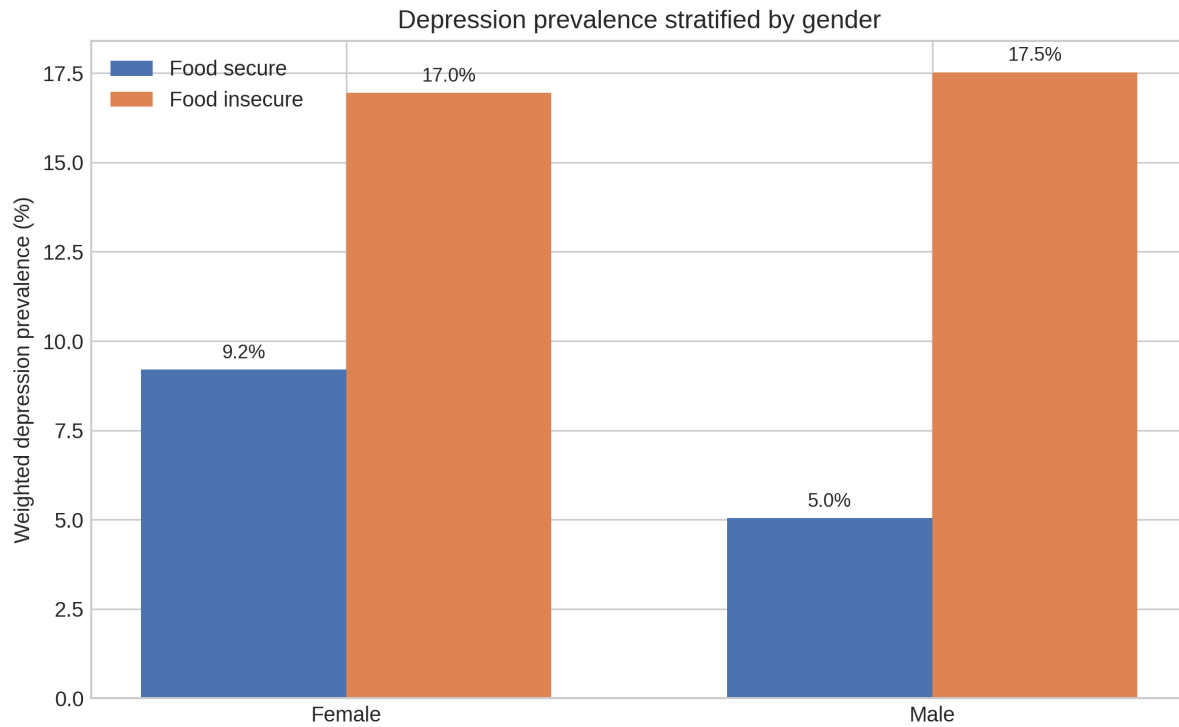


Figure 3. Weighted depression prevalence by food-security status, stratified by gender.

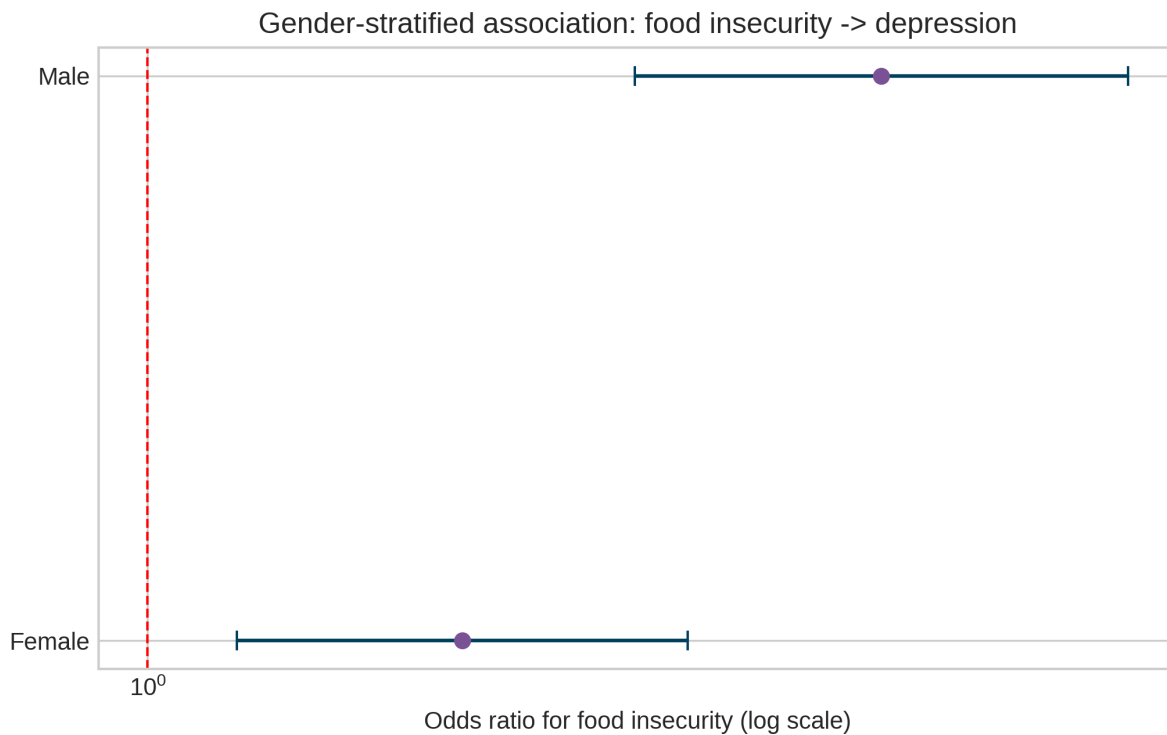


Figure 4. Gender-stratified adjusted odds ratios for the food insecurity and depression association.

5. Regression Results Tables

Table 1 presents adjusted associations from the pooled weighted logistic model.

Predictor	OR	95% CI	p-value
Age (years)	1.29	0.95 to 1.75	0.1070
BMI (kg/m ²)	1.04	1.02 to 1.06	0.0003
Food insecurity (vs secure)	3.04	2.13 to 4.34	<0.0001
Income-to-poverty ratio	1.11	1.01 to 1.23	0.0335
Male (vs female)	0.66	0.49 to 0.90	0.0085

Table 2 presents gender-stratified adjusted odds ratios for the food insecurity effect.

Gender	N	OR (Food insecurity)	95% CI	p-value
Female	1,007	1.97	1.21 to 3.21	0.0062
Male	1,074	4.87	2.86 to 8.30	<0.0001

6. Discussion and Public Health Implications

This analysis indicates a robust association between household food insecurity and depressive symptoms in U.S. youth. The relationship remains substantial after adjustment for age, sex, economic status, and BMI.

From a public health perspective, food access interventions may deliver dual benefits across nutritional and mental health domains. Potential policy channels include school meal optimization, targeted food assistance, and family-centered prevention strategies.

7. Limitations and Future Research

- Cross-sectional NHANES design limits causal interpretation.
- Complete-case analysis can introduce bias under non-random missingness.
- Residual confounding and measurement error remain possible.
- Future work should test mediation pathways and longitudinal relationships.

Reproducibility

All code, data-processing scripts, tables, and figures are available in the project repository: <https://github.com/bobaoxu2001/Food-insecurity>.